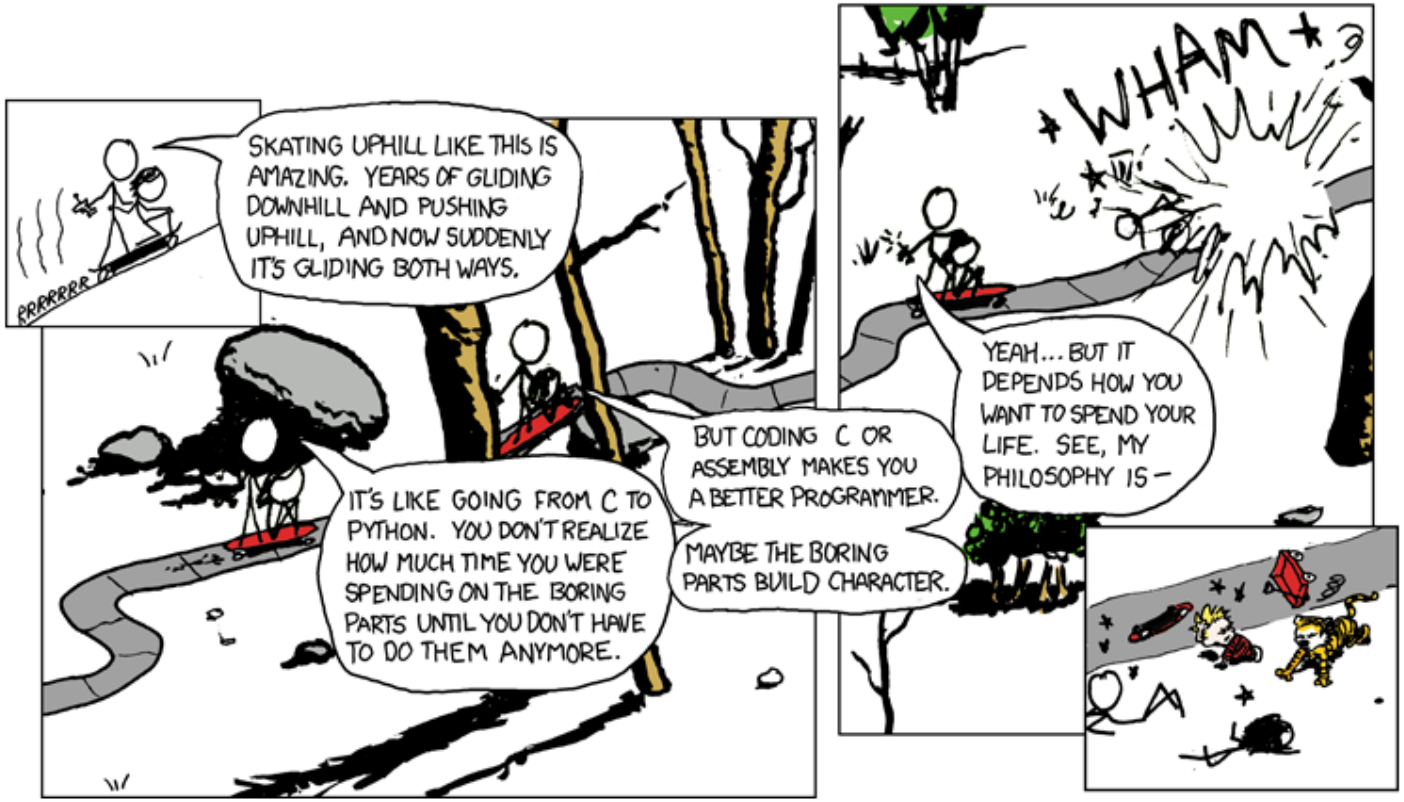
Introduction to Programming Languages and Techniques



xkcd.com

FULL PYTHON TUTORIAL

Full Python Tutorial

- Developed by Guido van Rossum in the early 1990s
- Named after Monty Python
- Available on eniac
- Available for download from http://www.python.org

Python

- ***Interpreted* language: work with an evaluator for language expressions (like DrJava, but more flexible)**
- ***Dynamically typed*: variables do not have a predefined type**
- **Rich, built-in collection types:**
  - Lists
  - Tuples
  - Dictionaries (maps)
  - Sets
- **Concise**

Language features

- ***Indentation instead of braces***
- **Several sequence types**
  - Strings `'…'` : made of characters, immutable
  - Lists `[…]` : made of anything, mutable
  - *Tuples* `(…)` *: made of anything, immutable*
- ***Powerful subscripting (slicing)***
- ***Functions are independent entities (not all functions are methods)***
- **Exceptions as in Java**
- **Simple object system**
- **Iterators (like Java) and *generators***

Why Python?

- **Good example of *scripting language***
- **“Pythonic” style is very concise**
- **Powerful but unobtrusive object system**
  - *Every* value is an object
- **Powerful collection and iteration abstractions**
  - Dynamic typing makes generics easy

Dynamic typing – the key difference

- **Java: *statically typed***
  - Variables are declared to refer to objects of a given type
  - Methods use type signatures to enforce contracts
- **Python**
  - Variables come into existence when first assigned to
  - A variable can refer to an object of any type
  - All types are (almost) treated the same way
  - *Main drawback*: type errors are only caught at runtime

Recommended Reading

- **On-line Python tutorials**
  - The Python Tutorial (http://docs.python.org/tutorial/)
    - Dense but more complete overview of the most important parts of the language
    - See course home page for others

- **PEP 8- Style Guide for Python Code**
  - http://www.python.org/dev/peps/pep-0008/
  - The official style guide to Python, contains many helpful programming tips

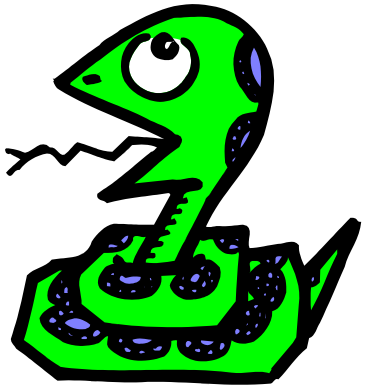
- ***Many* other books and on-line materials**
  - If you have a specific question, try Google first

IMPORTANT!

- **This slide deck is a *superset* of slides used in lecture.**
- **Extra slides have titles in Dark Red.**
- ***POINTS IN DARK RED ON THE SLIDES WILL ALSO BE SKIPPED IN LECTURE***
  - *Usually they're about parts of Python that are very much like Java*
- **SO I WON'T TALK ABOUT THIS POINT IN LECTURE**
- **The full slide set provides a reasonable manual for Python.**
- ***LEARN PYTHON BY PLAYING WITH EXAMPLES FROM THE SLIDES & MAKING UP YOUR OWN***
  - That Python is interpreted & simple makes this easy.....

Technical Issues

Installing & Running Python



Which Python?

- **Python 2.7**
  - Current version on Eniac, so we'll use it
  - Last stable release before version 3
  - Implements some of the new features in version 3, but fully backwards compatible
- **Python 3**
  - Released a few years ago
  - Many changes (including incompatible changes)
  - *Much* cleaner language in many ways
  - Strings use Unicode, not ASCII
  - But: A few important third party libraries are not yet compatible with Python 3 right now

The Python Interpreter

- **Interactive interface to Python**

```
% python
  Python 2.5 (r25:51908, May 25 2007, 16:14:04)
  [GCC 4.1.2 20061115 (prerelease) (SUSE Linux)] on linux2
  Type "help", "copyright", "credits" or "license" for more information.
  >>>
```

- **Python interpreter evaluates inputs:**

```
>>> 3*(7+2)
27
```

The IDLE GUI Environment (Windows)

Python Shell

File Edit Shell Debug Options Windows Help

```
Python 2.3.4 (#53, May 25 2004, 21:17:02) [MSC v.1200 32 bit (Intel)] on win32
Type "copyright", "credits" or "license()" for more information.

    ****************************************************************
    Personal firewall software may warn about the connection IDLE
    makes to its subprocess using this computer's internal loopback
    interface.  This connection is not visible on any external
    interface and no data is sent to or received from the Internet.
    ****************************************************************

IDLE 1.0.3
>>> for i in [x**2 for x in range(5)]:
        print i

0
1
4
9
16
>>> for i in [x**2 for x in range(
                                  range([start,] stop[, step]) -> list of integers
```

Ln: 20 Col: 34

IDLE Development Environment

- **Shell for interactive evaluation.**
- **Text editor with color-coding and smart indenting for creating Python files.**
- **Menu commands for changing system settings and running files.**

Running Interactively on UNIX (ENIAC)

On Unix…

```
% python

>>> 3+3
6
```

- **Python prompts with ‘>>>’.**
- **To exit Python (not Idle):**
  - In Unix, type CONTROL-D
  - In Windows, type CONTROL-Z + <Enter>

Running Programs on UNIX

% `python filename.py`

You can create python files using emacs.

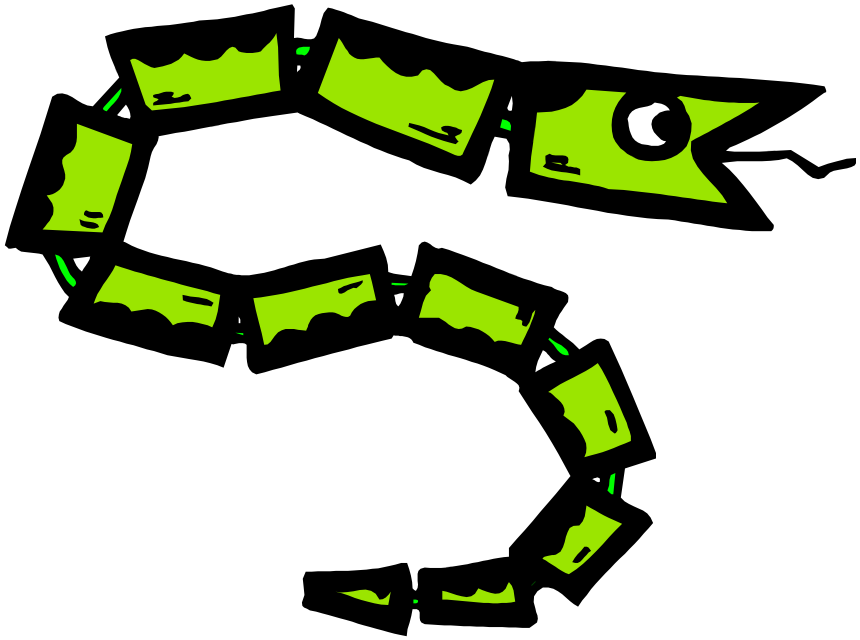
(There's a special Python editing mode.

`M-x python-mode`)

To make a python file executable, make this text the first line of the file :

#!/usr/bin/python

The Basics



A Code Sample (in IDLE)

```
x = 34 - 23                # A comment.
y = “Hello”                # Another one.
z = 3.45
if z == 3.45 or y == “Hello”:
    x = x + 1
    y = y + “ World”       # String concat.
print x
print y
```

Enough to Understand the Code

- **Indentation matters to the meaning of the code:**
  - Block structure indicated by indentation
- **The first assignment to a variable creates it.**
  - Variable types don't need to be declared.
  - Python figures out the variable types on its own.
- **Assignment uses `=` and comparison uses `==`.**
- **For numbers `+ - * / %` are as expected.**
  - Special use of `+` for string concatenation.
  - Special use of `%` for string formatting (as with printf in C)
- **Logical operators are words (`and, or, not`) *not* symbols**
- **Simple printing can be done with `print`.**

Basic Datatypes

- **Integers (default for numbers)**

```
z = 5 / 2    # Answer is 2, integer division.
```

- **Floats**

```
x = 3.456
```

- **Strings**
  - **Can use “” or ‘’ to specify.**
    “abc” ‘abc’ (Same thing.)
  - **Unmatched can occur within the string.**
    “matt’s”
  - **Use triple double-quotes for multi-line strings or strings than contain both ‘ and “ inside of them:**
    “““a‘b“c”””

Whitespace

Whitespace is meaningful in Python: especially indentation and placement of newlines.

- ***Use a newline to end a line of code.***
  - Use `\` when must go to next line prematurely.
- **No braces `{ }` to mark blocks of code in Python… *Use consistent indentation instead.***
  - *The first line with less indentation is outside of the block.*
  - *The first line with more indentation starts a nested block*
- **Often a colon appears at the start of a new block. (E.g. for function and class definitions.)**

Comments

- Start comments with # – the rest of line is ignored.
- Can include a "documentation string" as the first line of any new function or class that you define.
- The development environment, debugger, and other tools use it: it's good style to include one.

```
def my_function(x, y):
    """This is the docstring. This
    function does blah blah blah."""
    # The code would go here...
```

Assignment

- ***Binding a variable*** **in Python means setting a *name* to hold a *reference* to some *object*.**
  - *Assignment creates references, not copies (like Java)*
- **A variable is created *the first time* it appears on the left side of an assignment expression:**

```
x = 3
```

- **An object is deleted (by the garbage collector) once it becomes unreachable.**
- ***Names in Python do not have an intrinsic type. Objects have types.***
  - Python determines the type of the reference automatically based on what data is assigned to it.

(Multiple Assignment)

- **You can also assign to multiple names at the same time.**

```
>>> x, y = 2, 3
>>> x
2
>>> y
3
```

Naming Rules

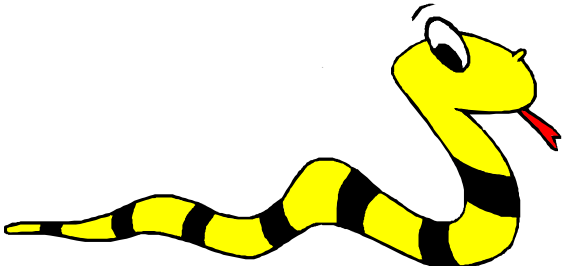
- **Names are case sensitive and cannot start with a number. They can contain letters, numbers, and underscores.**

  `bob  Bob  _bob  _2_bob_  bob_2  BoB`

- **There are some reserved words:**

  `and, assert, break, class, continue, def, del, elif, else, except, exec, finally, for, from, global, if, import, in, is, lambda, not, or, pass, print, raise, return, try, while`

Sequence types:
Tuples, Lists, and Strings



Sequence Types

1. Tuple
   - A simple ***immutable*** ordered sequence of items
     - Immutable: a tuple cannot be modified once created....
   - Items can be of mixed types, including collection types

2. Strings
   - ***Immutable***
   - Conceptually very much like a tuple
   - Regular strings use 8-bit characters. *Unicode strings* use 2-byte characters. (All this is changed in Python 3.)

3. List
   - ***Mutable*** ordered sequence of items of mixed types

Sequence Types 2

- **The three sequence types (tuples, strings, and lists) share much of the same syntax and functionality.**

- **Tuples are defined using parentheses (and commas).**

```
>>> tu = (23, 'abc', 4.56, (2,3), 'def')
```

- **Lists are defined using square brackets (and commas).**

```
>>> li = ["abc", 34, 4.34, 23]
```

- **Strings are defined using quotes (", ', or """").**

```
>>> st = "Hello World"
>>> st = 'Hello World'
>>> st = """This is a multi-line
string that uses triple quotes."""
```

Sequence Types 3

- **We can access individual members of a tuple, list, or string using square bracket "array" notation.**
- ***Note that all are 0 based…***

```
>>> tu = (23, 'abc', 4.56, (2,3), 'def')
>>> tu[1]      # Second item in the tuple.
 'abc'

>>> li = ["abc", 34, 4.34, 23]
>>> li[1]       # Second item in the list.
 34

>>> st = "Hello World"
>>> st[1]    # Second character in string.
 'e'
```

Negative indices

```
>>> t = (23, 'abc', 4.56, (2,3), 'def')
```

Positive index: count from the left, starting with 0.

```
>>> t[1]
'abc'
```

Negative lookup: count from right, starting with –1.

```
>>> t[-3]
4.56
```

Slicing: Return Copy of a Subset 1

```
>>> t = (23, 'abc', 4.56, (2,3), 'def')
```

Return a copy of the container with a subset of the original members. Start copying at the first index, and stop copying *before* the second index.

```
>>> t[1:4]
('abc', 4.56, (2,3))
```

You can also use negative indices when slicing.

```
>>> t[1:-1]
('abc', 4.56, (2,3))
```

Optional argument allows selection of every n^{th} item.

```
>>> t[1:-1:2]
('abc', (2,3))
```

Slicing: Return Copy of a Subset 2

```
>>> t = (23, 'abc', 4.56, (2,3), 'def')
```

Omit the first index to make a copy starting from the beginning of the container.

```
>>> t[:2]
(23, 'abc')
```

Omit the second index to make a copy starting at the first index and going to the end of the container.

```
>>> t[2:]
(4.56, (2,3), 'def')
```

Copying the Whole Sequence

To make a *copy* of an entire sequence, you can use `[:]`.

```
>>> t[:]
(23, 'abc', 4.56, (2,3), 'def')
```

Note the difference between these two lines for mutable sequences:

```
>>> list2 = list1     # 2 names refer to 1 ref
                      # Changing one affects both

>>> list2 = list1[:] # Two independent copies, two refs
```

The ‘in’ Operator

- **Boolean test whether a value is inside a collection (often called a *container* in Python:**

```
>>> t = [1, 2, 4, 5]
>>> 3 in t
False
>>> 4 in t
True
>>> 4 not in t
False
```

- **For strings, tests for substrings**

```
>>> a = 'abcde'
>>> 'c' in a
True
>>> 'cd' in a
True
>>> 'ac' in a
False
```

- **Be careful: the *in* keyword is also used in the syntax of *for loops* and *list comprehensions*.**

The + Operator

- **The + operator produces a *new* tuple, list, or string whose value is the concatenation of its arguments.**
- **Extends concatenation from strings to other types**

```
>>> (1, 2, 3) + (4, 5, 6)
 (1, 2, 3, 4, 5, 6)

>>> [1, 2, 3] + [4, 5, 6]
 [1, 2, 3, 4, 5, 6]

>>> “Hello” + “ ” + “World”
 ‘Hello World’
```

Mutability:
Tuples vs. Lists

Lists: Mutable

```
>>> li = ['abc', 23, 4.34, 23]
>>> li[1] = 45
>>> li
  ['abc', 45, 4.34, 23]
```

- **We can change lists *in place.***
- **Name *li* still points to the same memory reference when we're done.**

Tuples: Immutable

```
>>> t = (23, 'abc', 4.56, (2,3), 'def')
>>> t[2] = 3.14

Traceback (most recent call last):
  File "<pyshell#75>", line 1, in -toplevel-
    tu[2] = 3.14
TypeError: object doesn't support item assignment
```

You can't change a tuple.

You can make a fresh tuple and assign its reference to a previously used name.

```
>>> t = (23, 'abc', 3.14, (2,3), 'def')
```

- ***The immutability of tuples means they're faster than lists.***

Operations on Lists Only 1

```
>>> li = [1, 11, 3, 4, 5]

>>> li.append('a')  # Note the method syntax
>>> li
[1, 11, 3, 4, 5, 'a']

>>> li.insert(2, 'i')
>>>li
[1, 11, 'i', 3, 4, 5, 'a']
```

The *extend* method vs the + operator.

- **+ creates a fresh list (with a new memory reference)**
- ***extend* is just like *add* in Java; it operates on list `li` in place.**

```
>>> li.extend([9, 8, 7])
>>>li
[1, 2, 'i', 3, 4, 5, 'a', 9, 8, 7]
```

***Confusing*:**

- ***extend* takes a list as an argument *unlike Java***
- ***append* takes a singleton as an argument.**

```
>>> li.append([10, 11, 12])
>>> li
[1, 2, 'i', 3, 4, 5, 'a', 9, 8, 7, [10, 11, 12]]
```

Operations on Lists Only 3

```
>>> li = ['a', 'b', 'c', 'b']

>>> li.index('b')      # index of first occurrence*
1
```

*more complex forms exist

```
>>> li.count('b')      # number of occurrences
2

>>> li.remove('b')     # remove first occurrence
>>> li
  ['a', 'c', 'b']
```

Operations on Lists Only 4

```
>>> li = [5, 2, 6, 8]

>>> li.reverse()    # reverse the list *in place*
>>> li
  [8, 6, 2, 5]

>>> li.sort()       # sort the list *in place*
>>> li
  [2, 5, 6, 8]

>>> li.sort(some_function)
    # sort in place using user-defined comparison
```

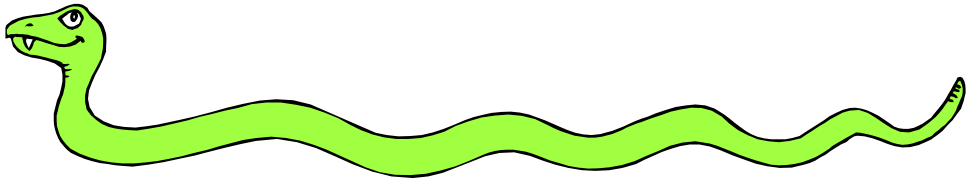
Tuples vs. Lists

- **Lists slower but more powerful than tuples.**
  - Lists can be modified, and they have lots of handy operations we can perform on them.
  - Tuples are immutable and have fewer features.

- **To convert between tuples and lists use the list() and tuple() functions:**

```
li = list(tu)
tu = tuple(li)
```

Dictionaries: a *mapping* collection type



Dictionaries: Like *maps* in Java

- **Dictionaries store a *mapping* between a set of keys and a set of values.**
  - ***Keys can be any immutable type.***
  - Values can be any type
  - Values and keys can be of different types in a single dictionary
- **You can**
  - define
  - modify
  - view
  - lookup
  - delete

  the key-value pairs in the dictionary.

Creating and accessing dictionaries

```
>>> d = {'user':'bozo', 'pswd':1234}

>>> d['user']
'bozo'

>>> d['pswd']
1234

>>> d['bozo']

Traceback (innermost last):
  File '<interactive input>' line 1, in ?
KeyError: bozo
```

Updating Dictionaries

```
>>> d = {'user':'bozo', 'pswd':1234}

>>> d['user'] = 'clown'
>>> d
{'user':'clown', 'pswd':1234}
```

- **Keys must be unique.**
- **Assigning to an existing key replaces its value.**

```
>>> d['id'] = 45
>>> d
{'user':'clown', 'id':45, 'pswd':1234}
```

- **Dictionaries are unordered**
  - **New entry might appear anywhere in the output.**
- **(Dictionaries work by *hashing*)**

Removing dictionary entries

```
>>> d = {'user':'bozo', 'p':1234, 'i':34}

>>> del d['user']              # Remove one. Note that del is
                               # a function.
>>> d
{'p':1234, 'i':34}

>>> d.clear()                  # Remove all.
>>> d
{}
>>> a=[1,2]
>>> del a[1]                  # (del also works on lists)
>>> a
[1]
```

Useful Accessor Methods

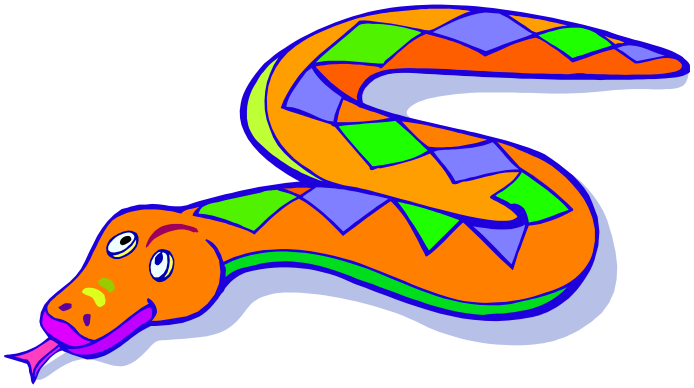
```
>>> d = {'user':'bozo', 'p':1234, 'i':34}

>>> d.keys()                # List of current keys
['user', 'p', 'i']

>>> d.values()              # List of current values.
['bozo', 1234, 34]

>>> d.items()         # List of item tuples.
[('user','bozo'), ('p',1234), ('i',34)]
```

Boolean Expressions



True and False

- ***True*** **and** ***False*** **are constants**

- ***Other values are treated as equivalent to either True or False when used in conditionals:***
  - *False*: zero, *None*, empty containers
  - *True*: non-zero numbers, non-empty objects
  - See PEP 8 for the most Pythonic ways to compare

- **Comparison operators: ==, !=, <, <=, etc.**
  - `X == Y`
    - X and Y have same value (like Java *equals* method)
  - `X is Y`:
    - X and Y refer to the *exact same object* (like Java ==)

Logical Operators

- **You can also combine Boolean expressions.**
  - *True* if a is True and b is True: `a and b`
  - *True* if a is True or b is True: `a or b`
  - *True* if a is False: `not a`

Conditional Expressions

- `x = true_value if condition else false_value`

- lazy evaluation:
  - **First, `condition` is evaluated**
  - **If *True*, `true_value` is evaluated and returned**
  - **If *False*, `false_value` is evaluated and returned**

Control Flow

if Statements (as expected)

```
if x == 3:
    print "X equals 3."
elif x == 2:
    print "X equals 2."
else:
    print "X equals something else."
print "This is outside the 'if'."
```

Note:

- **Use of indentation for blocks**
- **Colon (*:*) after boolean expression**

while Loops (as expected)

```
>>> x = 3
>>> while x < 5:
      print x, "still in the loop"
      x = x + 1
3 still in the loop
4 still in the loop
>>> x = 6
>>> while x < 5:
      print x, "still in the loop"

>>>
```

break and *continue*

- **You can use the keyword *break* inside a loop to leave the *while* loop entirely.**

- **You can use the keyword *continue* inside a loop to stop processing the current iteration of the loop and immediately go on to the next one.**

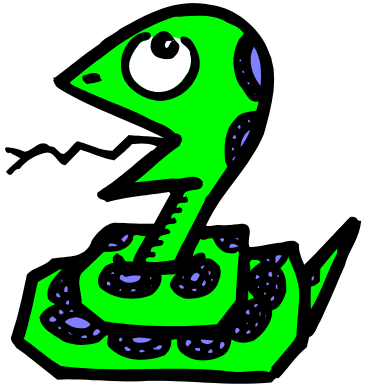
assert

- **An *assert* statement will check to make sure that something is true during the course of a program.**
  - If the condition if false, the program stops
    - (more accurately: throws an exception)

```
assert(number_of_players < 5)
```

- **Also found in Java; we just didn't mention it!**

For Loops



For Loops 1

- For-each is Python's *only* form of for loop
- A for loop steps through each of the items in a collection type, or any other type of object which is "iterable"

```
for <item> in <collection>:
  <statements>
```

- If <collection> is a list or a tuple, then the loop steps through each element of the sequence.

- If <collection> is a string, then the loop steps through each character of the string.

```
for someChar in "Hello World":
   print someChar
```

For Loops 2

```
for <item> in <collection>:
  <statements>
```

- **<item> can be more complex than a single variable name.**
  - If the elements of <collection> are themselves collections, then <item> can match the structure of the elements. (We saw something similar with list comprehensions and with ordinary assignments.)

```
for (x, y) in [(a,1), (b,2), (c,3), (d,4)]:
  print x
```

For loops and the *range()* function

- **We often want to write a loop where the variables ranges over some sequence of numbers. The *range()* function returns a list of numbers from 0 up to but not including the number we pass to it.**

- **range(5) returns [0,1,2,3,4]**
- **So we can say:**

```
for x in range(5):
    print x
```

- **(There are several other forms of *range()* that provide variants of this functionality…)**
- ***xrange()* returns an *iterator* that provides the same functionality more efficiently**

Abuse of the *range()* function

- **Don't use *range()* to iterate over a sequence solely to have the index and elements available at the same time**

- **Avoid:**

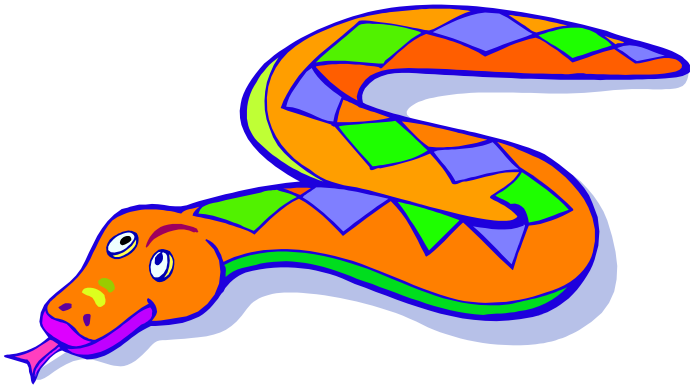
```
for i in range(len(mylist)):
    print i, mylist[i]
```

- **Instead:**

```
for (i, item) in enumerate(mylist):
    print i, item
```

- **This is an example of an *anti-pattern* in Python**
  - For more, see:
    - http://www.seas.upenn.edu/~lignos/py_antipatterns.html
    - http://stackoverflow.com/questions/576988/python-specific-antipatterns-and-bad-practices

Generating Lists using “List Comprehensions”



List Comprehensions 1

- **A powerful feature of the Python language.**
  - Generate a new list by applying a function to every member of an original list.
  - Python programmers use list comprehensions extensively. You'll see many of them in real code.

[expression for name in list]

[expression for name in list]

List Comprehensions 2

```
>>> li = [3, 6, 2, 7]
>>> [elem*2 for elem in li]
[6, 12, 4, 14]
```

[expression `for` name `in` list]

- Where expression is some calculation or operation acting upon the variable name.
- For each member of the list, the list comprehension
  1. sets name equal to that member, and
  2. calculates a new value using expression,
- It then collects these new values into a list which is the return value of the list comprehension.

[expression for name in list]

List Comprehensions 3

- **If the elements of list are other collections, then name can be replaced by a *collection* of names that match the "shape" of the list members.**

```
>>> li = [('a', 1), ('b', 2), ('c', 7)]
>>> [ n * 3 for (x, n) in li]
[3, 6, 21]
```

[expression for name in list if filter]

Filtered List Comprehension 1

- Filter determines whether expression is performed on each member of the list.

- When processing each element of list, first check if it satisfies the filter condition.

- If the filter condition returns *False*, that element is omitted from the list before the list comprehension is evaluated.

[expression for name in list if filter]

Filtered List Comprehension 2

```
>>> li = [3, 6, 2, 7, 1, 9]
>>> [elem * 2 for elem in li if elem > 4]
[12, 14, 18]
```

- **Only 6, 7, and 9 satisfy the filter condition.**
- **So, only 12, 14, and 18 are produced.**

Nested List Comprehensions

- **Since list comprehensions take a list as input and produce a list as output, they are easily nested:**

```
>>> li = [3, 2, 4, 1]
>>> [elem*2 for elem in
         [item+1 for item in li] ]
[8, 6, 10, 4]
```

- **The inner comprehension produces: [4, 3, 5, 2].**
- **So, the outer one produces: [8, 6, 10, 4].**

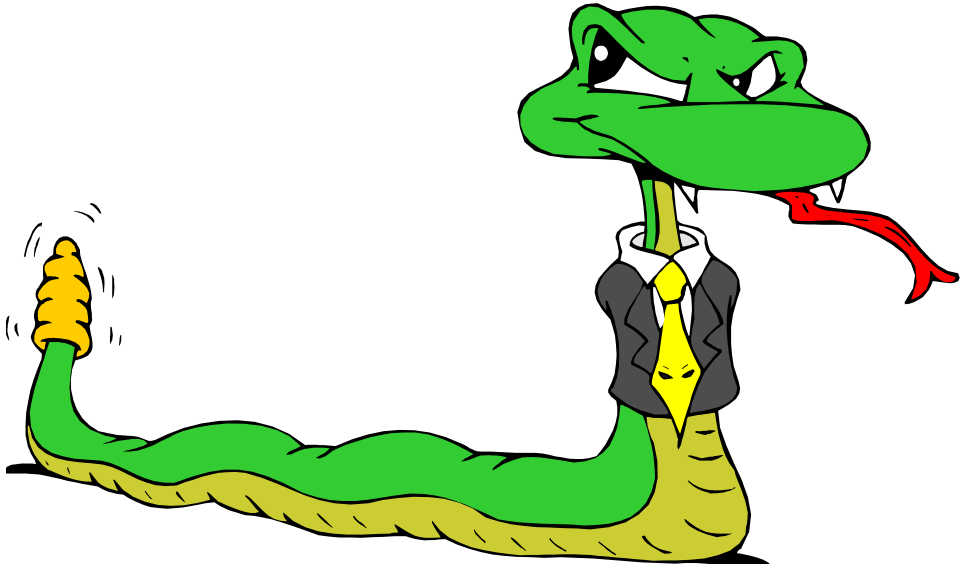
For Loops / List Comprehensions

- **Python's list comprehensions provide a natural idiom that usually requires a for-loop in other programming languages.**
  - As a result, Python code uses many fewer for-loops

- *Caveat*! **The keywords *for* and *in* also appear in the syntax of list comprehensions, but this is a totally different construction.**

Functions in Python

(Methods later)



Defining Functions

Function definition begins with **def**

Function name and its arguments.

```
def get_final_answer(filename):
    """Documentation String"""
    line1
    line2
    return total_counter
...
```

Colon.

First line with less indentation is considered to be outside of the function definition.

'return' indicates the value to be sent back to the caller.

No declaration of <u>types</u> of arguments or result

Calling a Function

```
>>> def myfun(x, y):
        return x * y
>>> myfun(3, 4)
12
```

Functions without returns

- ***All*** **functions in Python have a return value**
  - even if no *return* line inside the code.
- **Functions without a** *return* **return the special value** ***None*****.**
  - *None* is a special constant in the language.
  - *None* is used like *null* in Java.
  - *None* is also logically equivalent to False.
  - The interpreter doesn’t print *None*

Function overloading? No.

- **There is no function overloading in Python.**
  - ***Unlike Java, a Python function is specified by its name alone***
    - The number, order, names, or types of its arguments *cannot* be used to distinguish between two functions with the same name.
  - Two different functions can't have the same name, even if they have different numbers of arguments.

  - *But **operator** overloading – overloading +, ==, -, etc. – is possible using special methods on various classes (see later slides)*

Functions are first-class objects in Python

- Functions can be used just like any other data
- They can be
  - Arguments to function
  - Return values of functions
  - Assigned to variables
  - Parts of tuples, lists, etc
  - …

```
>>> def myfun(x):
        return x*3

>>> def apply(q, x):
        return q(x)

>>> apply(myfun, 7)
21
```

Lambda Notation

like anonymous
inner classes in Java

- **Functions can be defined without giving them names.**
- **This is most useful when passing a short function as an argument to another function.**

```
>>> apply(lambda z: z * 4, 7)
28
```

- **The first argument to apply() is an unnamed function that takes one input and returns the input multiplied by four.**
- **Note: only single-expression functions can be defined using this lambda notation.**
- **Lambda notation has a rich history in CS research and the design of many current programming languages.**

Default Values for Arguments

- You can provide **default** values for a function's arguments
- These arguments are optional when the function is called

```
>>> def myfun(b, c=3, d="hello"):
            return b + c
>>> myfun(5,3,"hello")
>>> myfun(5,3)
>>> myfun(5)
```

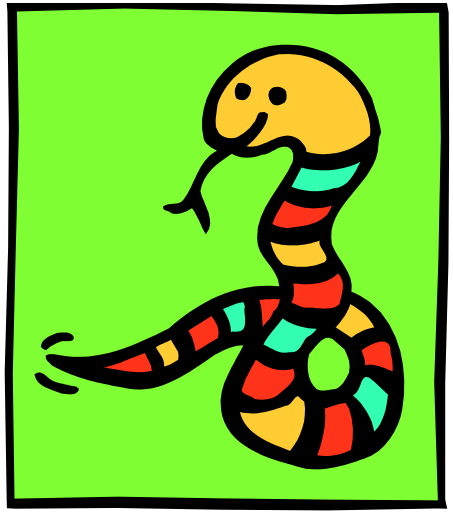
All of the above function calls return 8.

Keyword Arguments

- Functions can be called with arguments out of order
- These arguments are specified in the call
- Keyword arguments can be used for a final subset of the arguments.

```
>>> def myfun (a, b, c):
        return a-b
>>> myfun(2, 1, 43)
  1
>>> myfun(c=43, b=1, a=2)
  1
>>> myfun(2, c=43, b=1)
  1
```

Inheritance



Subclasses

- **A class can *extend* the definition of another class**
  - Allows use (or extension) of methods and attributes already defined in the previous one.
  - New class: *subclass*. Original: *parent*, *ancestor* or *superclass*

- **To define a subclass, put the name of the superclass in parentheses after the subclass's name on the first line of the definition.**
-

```
class ai_student(student):
```

  - Python has no 'extends' keyword like Java.
  - Multiple inheritance is supported.

Redefining Methods

- **Very similar to over-riding methods in Java**

- **To *redefine a method* of the parent class, include a new definition using the same name in the subclass.**
  - The old code won't get executed.

- **To execute the method in the parent class *in addition to* new code for some method, explicitly call the parent's version of the method.**

```
parentClass.methodName(self, a, b, c)
```

  - The only time you ever explicitly pass 'self' as an argument is when calling a method of an ancestor.

Extending __init__

- Very similar to Java
- Commonly, the ancestor's `__init__` method is executed in addition to new commands.
- *Must be done explicitly*
- You'll often see something like this in the `__init__` method of subclasses:

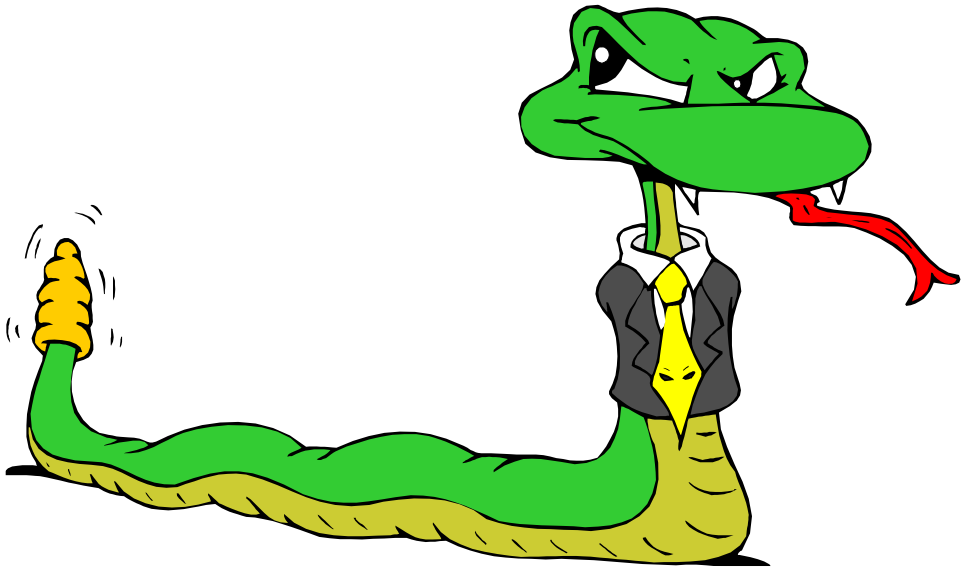
```
parentClass.__init__(self, x, y)
```

where parentClass is the name of the parent's class.

Private Data and Methods

- **Any attribute or method with two leading underscores in its name (but none at the end) is private. It cannot be accessed outside of that class.**
  - Note:
    Names with two underscores at the beginning ***and the end*** are for built-in methods or attributes for the class
  - Note:
    There is no ‘protected’ status in Python; so, subclasses would be unable to access these private data either

Importing and Modules



`Import` and Modules

- **Programs will often use classes & functions defined in another file**
- **A Python module is a single file with the same name (plus the *.py* extension)**
- ***Modules can contain many classes and functions***
- **Access using *import* (like Java)**

Where does Python look for module files?

- **The list of directories where Python looks: sys.path**
- **When Python starts up, this variable is initialized from the PYTHONPATH environment variable**
- **To add a directory of your own to this list, append it to this list.**

```
sys.path.append('/my/new/path')
```

  - **Oops! Operating system dependent….**

Import I

```
import somefile
```

- ***Everything*** **in somefile.py can be referred to by:**

```
somefile.className.method("abc")
somefile.myFunction(34)
```

- ```
  from somefile import *
  ```
  - ***Everything*** **in somefile.py can be referred to by:**

```
className.method("abc")
myFunction(34)
```

  - Careful! This can overwrite the definition of an existing function or variable!
```

Import II

```
from somefile import className
```

- **Only the item *_className_* in somefile.py gets imported.**
- **Refer to it without a module prefix.**
- ***Caveat*! This can overwrite an existing definition.**

`className.method("abc")` ← This was imported

~~`myFunction(34)`~~ ← Not this one

Commonly Used Modules

- **Some useful modules, included with Python:**

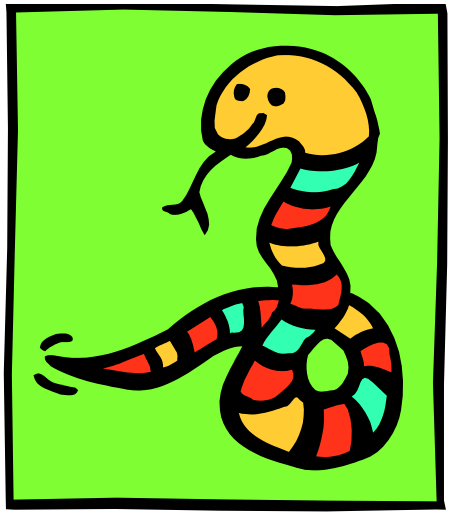
- **Module: sys - Lots of handy stuff.**
- **Module: os - OS specific code.**
- **Module: os.path - Directory processing.**

- **The Python standard library has *lots* of other useful stuff...**

More Commonly Used Modules

- **Module: math - Mathematical functions.**
  - Exponents
  - sqrt
- **Module: Random - Random numbers**
  - Randrange (good for simulations, games, …)
  - Uniform
  - Choice
  - Shuffle
- **To see what's in the standard library of modules, check out the Python Library Reference:**
  - http://docs.python.org/lib/lib.html

String Operations



String Operations

- **The string class provides a number of methods for useful formatting operations:**

```
>>> "hello".upper()
'HELLO'
```

- **Check the Python documentation for many other handy string operations.**
- **Helpful hint: use `<string>.strip()` to strip off final newlines from lines read from files**

String Formatting Operator: *%*

- **The operator *%* allows strings to be built out of many data items in a "fill in the blanks" fashion.**
  - Allows control of how the final string output will appear.
  - For example, we could force a number to display with a specific number of digits after the decimal point.

- **Very similar to the sprintf command of C.**

```
>>> x = "abc"
>>> y = 34
>>> "%s xyz %d" % (x, y)
'abc xyz 34'
```

- **The tuple following the *%* operator is used to fill in the blanks in the original string marked with *%s* or *%d*.**
  - Check Python documentation for details.

Printing with Python

- print a string to the standard output stream using "print"
- Using the % string operator in combination with the print command, we can format our output text.

```
>>> print  "%s xyz %d"  %  ("abc", 34)
abc xyz 34
```

"Print" automatically adds a newline to the end of the string. If you include a list of strings, it will concatenate them with a space between them.

```
>>> print "abc"
abc
```

```
>>> print "abc", "def"
abc def
```

- Useful trick: `>>> print "abc",` doesn't add newline (does add space)

String to List to String

- **Join turns a list of strings into one string.**

  <separator_string>.join(<some_list>)

```
>>> ";".join( ["abc", "def", "ghi"] )
"abc;def;ghi"
```

- **Split turns one string into a list of strings.**

  <some_string>.split(<separator_string>)

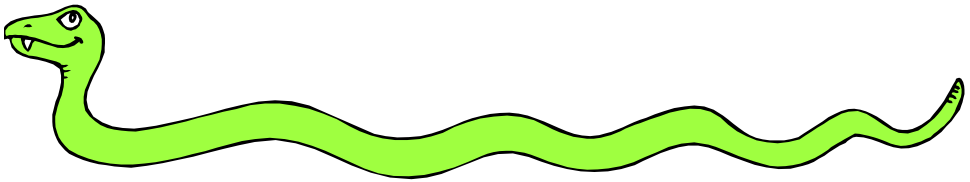
```
>>> "abc;def;ghi".split( ";" )
["abc", "def", "ghi"]
```

Convert Anything to a String

- **The built-in str() function can convert an instance of <u>any</u> data type into a string.**
  - You can define how this function behaves for user-created data types. You can also redefine the behavior of this function for many types.

```
>>> "Hello " + str(2)
"Hello 2"
```

Special Built-In Methods and Attributes



Built-In Members of Classes

- **Classes contain many methods and attributes that are included by Python even if you don't define them explicitly.**
  - Most of these methods define automatic functionality triggered by special operators or usage of that class.
  - The built-in attributes define information that must be stored for all classes.
- **All built-in members have double underscores around their names: `__doc__`**

Special Methods

- **For example, the method `__repr__` exists for all classes, and you can always redefine it.**
- **The definition of this method specifies how to turn an instance of the class into a string.**
  - `print f` sometimes calls `f.__repr__()` to produce a string for object f.
  - If you type `f` at the prompt and hit ENTER, then you are also calling `__repr__` to determine what to display to the user as output.

Special Methods – Example

```
class student:
    ...
     def __repr__(self):
       return "I'm named " + self.full_name
    ...

>>> f = student("Bob Smith", 23)
>>> print f
I'm named Bob Smith
>>> f
"I'm named Bob Smith"
```

Special Methods

- **Used to implement operator overloading**
  - Most operators trigger a special method, dependent on class

`__init__` :	The constructor for the class.
`__len__` :	Define how **`len(`** obj **`)`** works.
`__copy__` :	Define how to copy a class.
`__cmp__` :	Define how **`==`** works for class.
`__add__` :	Define how **`+`** works for class
`__neg__` :	Define how unary negation works for class

- **Other built-in methods allow you to give a class the ability to use [ ] notation like an array or ( ) notation like a function call.**

Special Data Attributes

- **These attributes exist for all classes.**

  `__doc__` : Variable storing the documentation string for that class.

  `__class__` : Variable which gives you a reference to the class from any instance of it.

  `__module__` : Variable which gives you a reference to the module in which the particular class is defined.

  __dict__ :The dictionary that is actually the namespace for a class (but not its superclasses).

- **Useful:**
  - **`dir(x)` returns a list of all methods and attributes defined for object x**

Special Data Items – Example

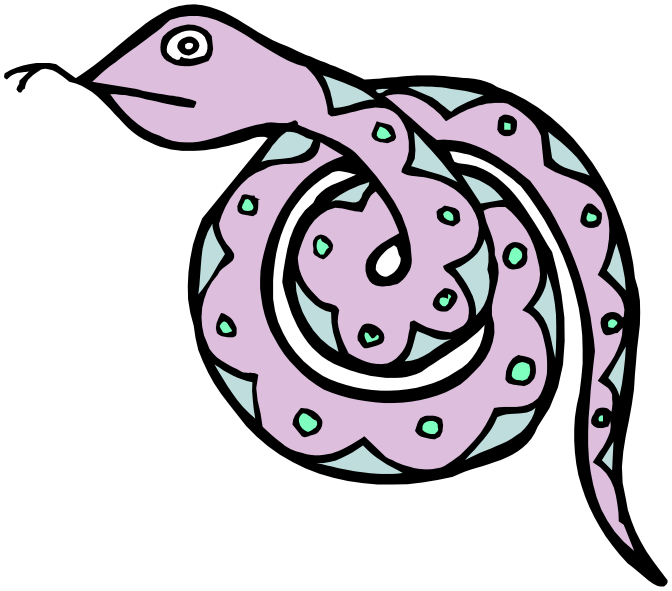
```
>>> f = student("Bob Smith", 23)
```

```
>>> print f.__doc__
A class representing a student.

>>> f.__class__
< class studentClass at 010B4C6 >

>>> g = f.__class__("Tom Jones", 34)
```

File Processing, Error Handling



File Processing with Python

About what you'd expect….

```
fileptr = open("filename")
somestring = fileptr.read()
for line in fileptr:
   print line
fileptr.close()
```

Exception Handling

- **Exceptions are Python objects**
  - More specific kinds of errors are subclasses of the general Error class.

- **You use the following forms to interact with them:**
  - try
  - except
  - *else*
  - finally

for example...

```
>>> def divide(x, y):
        try:
            result = x / y
        except ZeroDivisionError:
            print "division by zero!"
        else:
            print "result is", result
        finally:
            print "executing finally clause"

>>> divide(2, 1)
result is 2
executing finally clause
>>> divide(2, 0)
division by zero!
executing finally clause
>>> divide("2", "1")
executing finally clause
Traceback (most recent call last):
  File "<stdin>", line 1, in ?
  File "<stdin>", line 3, in divide
TypeError: unsupported operand type(s) for /: 'str' and 'str'
```

Iterators

Iterators in Python

```
>>> for e in [1,2]:
     print e
1
2
>>> s = [1,2]
>>> it = iter(s)
>>> it
<iterator object at 0x00A1DB50>
>>> it.next()
1
>>> it.next()
2
>>> it.next()
(most recent call last):
  File "<stdin>", line 1, in ?
    it.next()
StopIteration
```

Class with iterator in Python

```
class Reverse:
    "Iterator for looping over a sequence backwards"
    def __init__(self, data):
        self.data = data
        self.index = len(data)
    def next(self):
        if self.index == 0:
            raise StopIteration
        self.index = self.index - 1
        return self.data[self.index]
    def __iter__(self):
        return self
>>> for char in Reverse('spam'):
      print char
m
a
p
s
```

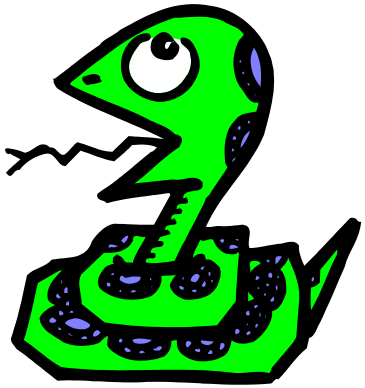
An iterator is any object with a "next" method

Iterators and list comprehensions

```
>>> [x for x in Reverse('spam')]
['m', 'a', 'p', 's']

>>> [x + x for x in Reverse('spam')]
['mm', 'aa', 'pp', 'ss']
```

Generators



Generators

- Defines an iterator with a function
- Maintains local state automatically

```
def reverse(data):
    for i in range(len(data)):
        yield data[len(data)-1-i]

>>> for char in reverse('spam'):
       print char
m
a
p
s
```

Using generators

- **Merging sequences:**

```
def merge(l, r):
  llen, rlen, i, j = len(l), len(r), 0, 0
  while i < llen or j < rlen:
    if j == rlen or (i < llen and l[i] < r[j]):
      yield l[i]
      i += 1
    else:
      yield r[j]
      j += 1
```

Using generators

```
>>> g = merge([2,4], [1, 3, 5])
>>> while True:
        print g.next()
1
2
3
4
5

Traceback (most recent call last):
  File "<pyshell#73>", line 2, in <module>
    print g.next()
StopIteration

>>> [x for x in merge([1,3,5],[2,4])]
[1, 2, 3, 4, 5]
```

Generators and exceptions

```
>>> g = merge([2,4], [1, 3, 5])
>>> while True:
      try:
        print g.next()
      except StopIteration:
        print 'Done'
        break
1
2
3
4
5
Done
```

(expression for name in list if filter)

List *Generators*

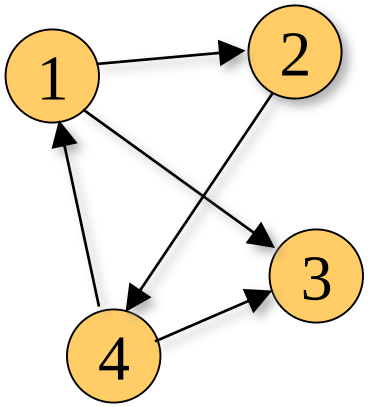
```
>>> a = (x * x for x in xrange(5))

>>> a
>>> <generator object <genexpr> at 0x031A7A80>

>>> for x in a:
    ...:     print x
    ...:
0
1
4
9
16
```

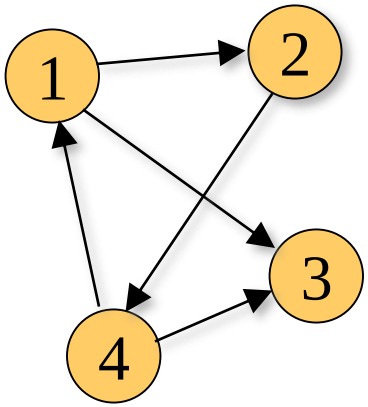
A directed graph class

```
>>> d = DiGraph([(1,2),(1,3),(2,4),(4,3),(4,1)])
>>> print d
1 -> 2
1 -> 3
2 -> 4
4 -> 3
4 -> 1
```



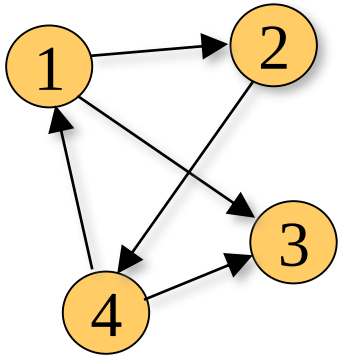
A directed graph class

```
>>> d = DiGraph([(1,2),(1,3),(2,4),(4,3),(4,1)])
>>> [v for v in d.search(1)]
[1, 2, 4, 3]
>>> [v for v in d.search(4)]
[4, 3, 1, 2]
>>> [v for v in d.search(2)]
[2, 4, 3, 1]
>>> [v for v in d.search(3)]
[3]
```



`search` method returns a *generator* for the nodes that can be reached from a given node by following arrows "from tail to head"

The DiGraph constructor



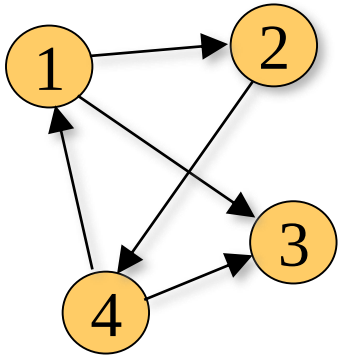

```
class DiGraph:
  def __init__(self, edges):
    self.adj = {}
    for u,v in edges:
        if u not in self.adj: self.adj[u] = [v]
        else: self.adj[u].append(v)

  def __str__(self):
    return '\n'.join(['%s -> %s'%(u,v) \
                      for u in self.adj for v in self.adj[u]])
  ...

>>> d = DiGraph([(1,2),(1,3),(2,4),(4,3),(4,1)])
>>> [v for v in d.search(1)]
{1: [2, 3], 2: [4], 4: [3, 1]}
```

The constructor builds a dictionary (`self.adj`) mapping each node name to a list of node names that can be reached by following one edge (an "adjacency list")

The `search` method




```
class DiGraph:
  ...

  def search(self, u, visited=set()):
    # If we haven't already visited this node...
    if u not in visited:
      # yield it
      yield u
      # and remember we've visited it now.
      visited.add(u)
      # Then, if there are any adjacant nodes...
      if u in self.adj:
        # for each adjacent node...
        for v in self.adj[u]:
          # search for all nodes reachable from *it*...
          for w in self.search(v, visited):
            # and yield each one.
            yield w
```